

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q21-1941

Pittsfield Generating Lp 100 kV
March 01, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1941. The Interconnection Customer proposes a 63.0 MW solar photovoltaic facility interconnecting at the Pittsfield Generating Lp 100 kV substation in the Ohio area. The request was received on November 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 625 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$6,593,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Mariposa to Msher 100.0 kV	Barrett Substation to Norich 500.0 kV Line	130.1	179	6.4
Mariposa to Acec Brooks 100.0 kV L	Barrett to Norwich 500.0 kV	124.6	205	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mariposa to Mosher 100.0 kV	Rebuild line with higher-capacity conductor	\$2,736,000	100% (\$2,736,000)
Mariposa to Acec Brooks 100.0 kV L	Rebuild line with higher-capacity conductor	\$3,857,000	100% (\$3,857,000)
	Total allocated to project		\$6,593,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.